Ain Shams University

Faculty of Engineering

Department of Computer & Systems Engineering

CSE476s (UG2018) – Fundamentals of Big-Data Analytics

Healthcare Provider Data Intelligence Pipeline

Web Crawling, Data Analysis & Visualization of Vezeeta's Provider Directory

Target: https://www.vezeeta.com/

1,640+	13,000+	3
Pages to Crawl	Doctor Records	Analysis Dimensions

Phase 1.0 – Idea Selection Report

1. Team Members

#	STUDENT NAME	STUDENT ID
1	Amr Ahmed Wahidi	2200429
2	Omar Mohamed Hassan	2200294
3	Youssef Emadeldeen Disoky	2200536
4	Hamza Basem Mohamed Ahmed	2200496
5	Belal Ahmed Eid	2200567
6	Youssef Ashraf Mohammed	2201056
7	Anas Ayman Mohammed	2200325
8	Ali Ahmed Mohamed Abdelmaboud	2200927
9	Abdallah Khaled Allam	2200969
10	Ammar Mohamed Abdelghany	2200707

2. Problem Description

Vezeeta (vezeeta.com) is the **largest digital healthcare platform in the Middle East and North Africa**, connecting millions of patients with thousands of doctors across Egypt and the region. The platform has fundamentally transformed how patients discover and book medical appointments — offering doctor profiles, specialty search, patient reviews, consultation fees, clinic locations, and real-time appointment availability all in one place.

Vezeeta's provider directory spans over **1,640 pages** of listings, with approximately **8 physician profiles per page**, totaling an estimated **13,000+ doctor records**. This represents one of the most comprehensive healthcare provider databases in Egypt. However, while this data is publicly displayed across thousands of individual pages, it has never been aggregated, structured, or analyzed as a unified dataset.

This means **critical business intelligence** that could drive Vezeeta's growth, improve patient experience, and optimize market strategy remains completely untapped:

- ✓ Which medical specialties are underserved in specific neighborhoods or governorates?
- ✓ What is the average consultation fee range per area or specialty — and where are pricing anomalies?
- ✓ Which geographic zones have no available appointments during peak or evening hours?
- ✓ How does physician density compare across high-value regions (e.g., Smoha vs. Maadi vs. Degla)?
- ✓ Where should Vezeeta focus recruitment efforts to fill coverage gaps and increase bookings?

> **Core Challenge:** Vezeeta sits on a goldmine of healthcare market data — but without structured extraction and analysis, this data cannot be leveraged for strategic decision-making, revenue optimization, or competitive advantage. This project transforms raw directory listings into a powerful business intelligence asset.

Platform Data Dimensions:

13,000+ Doctors
Largest provider network in the region.

Multiple Specializations
Dozens of medical specialties covered.

Nationwide Coverage
Governorates, cities & neighborhoods.

Consultation Fees
Varied pricing across regions & specialties.

Ratings & Reviews
Patient satisfaction & trust signals.

Working Hours
Availability patterns & time coverage.

3. Business Benefits

Imagine presenting these findings directly to Vezeeta's leadership team. Here is exactly how this data intelligence platform translates into **revenue growth, market expansion, and competitive advantage**:

$ 3.1 Direct Revenue Growth Opportunities

- **Fill coverage gaps = more bookings:** By identifying areas where specific specialties are missing or underserved, Vezeeta can recruit doctors to fill those gaps — every new doctor in an underserved area means new bookings and new revenue. 💰 **Revenue**
- **Premium placement for high-demand areas:** Data reveals which specialties patients search for most in each region. Vezeeta can offer premium listing placements to doctors in high-demand zones — a direct advertising revenue stream. 💰 **Ad Revenue**
- **Dynamic pricing intelligence:** Understanding fee distributions per area allows Vezeeta to advise doctors on competitive pricing — doctors who price right get more bookings, and more bookings mean more platform commission. 💰 **Commission**
- **Evening/night coverage monetization:** Identifying time-based dead zones (e.g., no doctors available after 9 PM) reveals an unmet demand. Vezeeta can incentivize doctors to extend hours in these windows — capturing patients who currently have no options. 💰 **New Market**

3.2 Strategic Market Expansion

- **Data-driven expansion planning:** Instead of guessing where to expand, Vezeeta can use geographic density analysis to identify high-population areas with low doctor coverage — these are the highest-ROI expansion targets. 💰 **Growth**
- **Specialty recruitment strategy:** If Dermatology has 200 doctors in Cairo but only 5 in Mansoura, that's a clear recruitment priority. Data makes recruitment surgical, not random. 💰 **Efficiency**
- **Targeted advertising campaigns:** Knowing exactly which areas have the highest patient demand but lowest doctor supply enables hyper-targeted ads — higher conversion rates, lower cost per acquisition. 💰 **Marketing ROI**
- **Partnership intelligence:** Hospitals and clinic chains can be pitched with data showing exactly where they should open new branches — Vezeeta becomes not just a booking platform but a strategic advisor. 💰 **B2B Revenue**

3. Business Benefits

3.3 Competitive Advantage & Platform Intelligence

- **Market intelligence moat:** No competitor has this level of structured, analyzed provider data. This becomes a proprietary intelligence asset that strengthens Vezeeta's market leadership.
- **Smarter recommendation engine:** Analysis of ratings, fees, availability, and location enables better doctor matching — improving patient satisfaction and repeat bookings.
- **Investor-ready analytics:** Structured market data (coverage maps, growth trends, pricing indices) is exactly what investors want to see — demonstrating deep market understanding.

3.4 Patient Experience = Platform Loyalty = Revenue

- **Faster doctor discovery:** Patients currently browse hundreds of pages. Structured data enables intelligent filtering by specialty, area, price, and availability — reducing search time from minutes to seconds. Happy patients return. 💰 **Retention**
- **Pricing transparency builds trust:** When patients can compare fees across doctors and areas, they trust the platform more — trust drives repeat usage and word-of-mouth growth. 💰 **Organic Growth**
- **Availability awareness reduces frustration:** Showing patients where and when doctors are actually available prevents wasted clicks and abandoned searches — improving conversion rates. 💰 **Conversion**
- **Quality signals drive premium bookings:** Correlating ratings with fees and experience helps surface the best doctors — patients willing to pay more for quality generate higher per-booking revenue. 💰 **ARPU**

4. Analysis Dimensions

The project defines three core analysis dimensions, each targeting a specific aspect of the healthcare market:

4.1 Geographic Distribution Analysis

- **Doctor Count:** Number of doctors per governorate / region.
- **Specialty Density:** Count of each specialty within every specific area.
- **Gap Identification:** Identifying regions with a shortage of specific specialties.
- **Market Saturation:** Pinpointing oversaturated areas with excess supply.

Goal: Map out Geographic Gaps and Saturation points to optimize expansion.

4.2 Pricing Analysis

- **Consultation Benchmarks:** Average consultation fee per specialty.
- **Regional Pricing:** Average price indexed by geographic location.
- **Comparative Analysis:** Comparing fees for the same specialty across different regions.
- **Market Positioning:** Detecting overpricing or underpricing relative to area averages.
- **Value Correlation:** Analyzing the relationship between price and patient ratings.

Goal: Understand Pricing Patterns to ensure competitive and logical fee structures.

4.3 Time Availability Analysis (Operating Hours)

- **Hourly Coverage:** Number of available doctors for every hour of the day.
- **Availability Distribution:** How working hours are spread throughout the day.
- **Regional Availability:** Comparing operating hours across different locations.
- **Peak & Off-Peak:** Identifying peak hours vs. underserved low-coverage hours.
- **Cross-Variable:** Working hours vs. Price, Working hours vs. Ratings, Region vs. Avg hours.

Goal: Optimize Time Coverage, maximize revenue, and enhance User Experience.

5. KPIs & Planned Key Insights

The following KPIs and insights will be the primary deliverables of the analysis phase:

INSIGHT	DESCRIPTION	BUSINESS VALUE
Specialty Gap Map	Which specialties are missing in each area	Identifies underserved communities & recruitment targets
Price Index by Area	Average fees per neighborhood & specialty	Pricing transparency & competitive benchmarking
Appointment Deserts	Areas with zero available bookings in a specialty	Supports resource allocation & expansion decisions
Doctor Density Score	Physicians per estimated population zone	Geographic equity analysis & market opportunity
Top Shortage Alerts	3 most critical specialty-area gaps	Actionable priorities for recruitment & growth
Peak Hour Coverage	Doctor availability per hour of day	Identifies time-based service gaps & revenue windows
Price vs. Rating	Correlation between fees and patient satisfaction	Pricing strategy & quality optimization

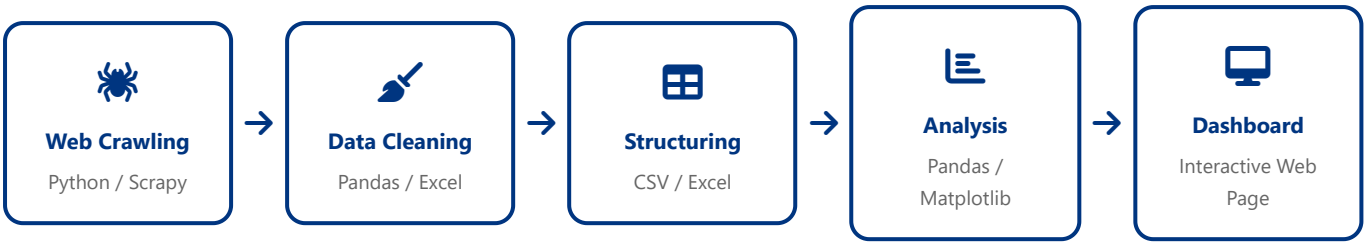
5. KPIs & Planned Key Insights

Target Data Schema:

Each physician record in the master dataset will capture the following fields:

FIELD	DESCRIPTION	EXAMPLE VALUE
Doctor Name	Full name of the physician	Dr. Ahmed Mohamed Khalil
Specialty	Primary medical specialty	Cardiology
Area / Neighborhood	Geographic location of the clinic	Smoha, Alexandria
Consultation Fee	Price range for a consultation	200–350 EGP
Available Hours	Working hours / appointment slots	Sat–Wed, 5–9 PM
Rating / Reviews	Patient review score or wait time	4.2 stars / 15 min avg

6. Technical Approach



6.1 Data Extraction

A web scraping pipeline will traverse all 1,640+ pages of Vezeeta's provider directory. The scraper will extract structured fields per physician record and output data to a master file. Key considerations include handling pagination, rate-limiting to avoid being blocked, and gracefully managing missing or inconsistently formatted fields.

6.2 Data Cleaning & Structuring

Raw extracted data will be cleaned and normalized. This includes standardizing specialty names, unifying geographic area labels, converting fee ranges to numeric form, flagging records with missing critical fields, and checking for duplicate records and outliers.

6.3 Analysis & Insight Generation

The clean dataset will be analyzed across the three defined dimensions: Geographic Distribution, Pricing Patterns, and Time Availability. Outputs include distribution maps, fee summaries, availability reports, and shortage alerts.

6.4 Visualization & Dashboard

An interactive web page will be built to browse the collected data and display analytical charts, filters, and geographic visualizations — making the data useful and accessible.

6.5 AI / LLM Features (Optional)

The team may explore AI-assisted tools for data interpretation, text classification, or intelligent search. The primary deliverable remains the structured dataset and analysis — AI enhancement is optional and may be introduced in a later phase.

7. Project Timeline

A clear month-by-month plan for developing the project:

Feb — Phase 1.0
- Problem Description
- Business Benefits
- Define KPIs
- Data Collection Plan

Mar — Phase 2
- Web Crawling (1,640 pages)
- Structured Initial Dataset
- Data Cleaning
- *27 Mar – Midterm Exams*

Apr — Phase 3
- Data Cleaning (Cont.)
- Data Preparation
- Analysis (3 Dimensions)
- Insights & Findings

May — Phase 4
- Interactive Dashboard
- AI / LLM Features (Optional)
- Final Report

Final — Exams
- Final Submission
- Presentation
- Demo Video

Project Scope: The project is scoped to physician data only — no pharmaceuticals, medical equipment, or hospital facilities. The dataset is expected to contain approximately 13,000–14,000 physician records across multiple specialties and geographic areas in Egypt. Out of scope: integration with external booking systems, real-time data updates, and patient-facing application development.

8. Ethical and Academic Considerations

- Only publicly accessible data from Vezeeta will be collected — no login-required or restricted data.
- No personal, private, or sensitive patient information will be stored or processed.
- The web crawler will respect rate limits and the website's `robots.txt` directives.
- Crawling will be performed with reasonable delays to avoid any disruption to Vezeeta's services.
- All collected data will be used exclusively for academic purposes under the CSE476s course.
- The project adheres to Ain Shams University's academic integrity and ethical research guidelines.
- No data will be redistributed, sold, or used for any commercial purpose.

> **Disclaimer:** This project is developed purely for educational and academic purposes as part of the CSE476s – Fundamentals of Big-Data Analytics course at Ain Shams University, Faculty of Engineering. The team does not claim any ownership of the data available on Vezeeta.com.

Risks & Mitigations:

RISK	LIKELIHOOD	IMPACT	MITIGATION
Website blocks scraper IP	Medium	High	Implement rate limiting, use rotating requests
Inconsistent data format across pages	High	Medium	Build flexible parser; manual review for edge cases
Team bandwidth during exam season	High	High	Front-load heavy tasks; use modular work distribution
Missing fields (fees, hours)	Medium	Medium	Flag as null; partial records still useful for analysis
Scope creep (AI models, UI)	Low	Medium	Defer to later phase; current scope is data + analysis only